
Intervention-Aware Spatiotemporal Point Process Modeling for Predictive Multi-Robot Bird Deterrence in Vineyards

Gabriel Basus

Proposed MS Thesis

Advisor: Siavash Farzan



Thesis Objectives & Research Question

Goals:

- Develop an intervention-aware online spatiotemporal intensity model by extending the current self-exciting formulation with an inhibitory term driven by deterrence actions.
- Integrate the new model into the existing task pipeline by scoring candidate patrol and deterrence locations using expected future intensity reduction per unit cost
- Evaluate performance in simulation against strong baselines (reactive and prediction-only) using metrics aligned with agricultural impact, energy, and coordination overhead

Research Question: can an online spatiotemporal point-process model both predict hotspots and represent suppression from deterrence actions so task selection reduces future bird activity and value-weighted exposure better than reactive or prediction-only baselines?

Expected Contributions

- Practical online intervention-aware spatiotemporal point process model suitable for real-time robotics
- A planning objective that selects action based on predicted future reduction, not only current hotspots
- A decentralized coordination mechanism that shares both detections and intervention effects locally
- An evaluation against reactive and prediction-only baselines using metrics tied to agricultural impact, energy, and communication cost

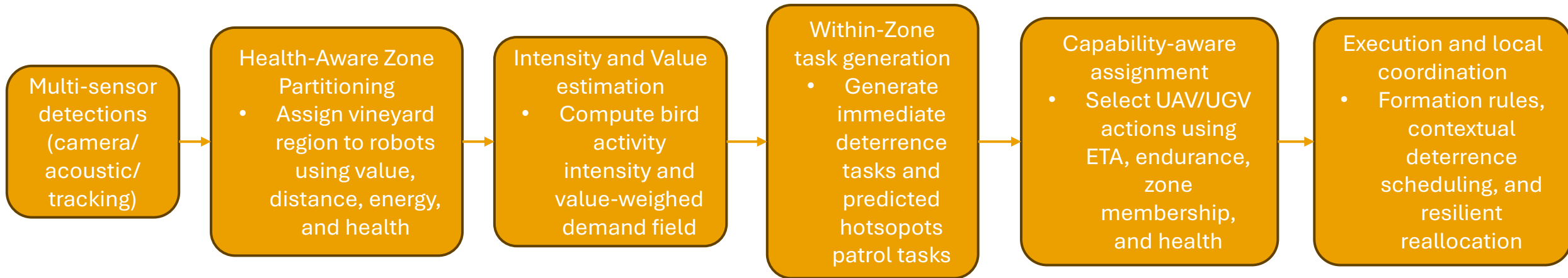
Why this Problem Matters

- Bird activity is bursty and spatially concentrated
- Static deterrents lose effectiveness over time due to habituation
- Growers need adaptive, mobile, real-time response in row-structured vineyards
- Static deterrents lose effectiveness due to habituation, and growers need adaptive response

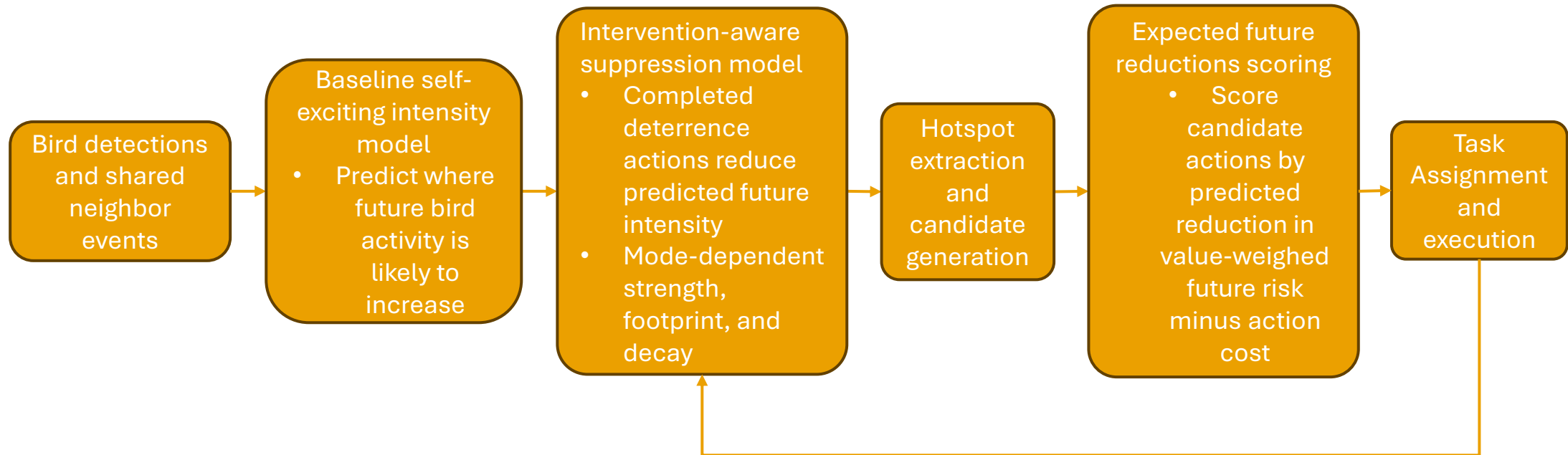
Limitations of Existing Approaches

- **Netting:** often the most reliable physical exclusion method, but it can carry substantial seasonal deployment, removal, storage, and equipment costs. One BC government guide gives an example of about \$2,500 per acre for nets plus about \$1,200 for placement/removal equipment for one setup.
- **Acoustic and visual scare tactics:** sound cannons, distress calls, and visual repellents can help, but birds habituate, so these methods work best when moved, rotated, or varied frequently.
- **Fixed deterrence schedules:** many systems still rely on static or manually tuned cue rotations, which do not adapt well to changing activity patterns or recent deterrence history. Your thesis explicitly targets that gap by moving to an intervention-aware model in which completed deterrence actions suppress future predicted risk, and your current implementation now preserves mode-specific suppression over time instead of collapsing all interventions into one inhibition state.

Current System Overview



Proposed Intervention-Aware Model



Model and Planning Contribution

Technical Contribution

- Extend the current self-exciting spatiotemporal intensity model with an inhibitory intervention term driven by completed deterrence actions.
- Model suppression using parameters that control:
 - Immediate deterrence strength
 - Persistence over time
 - Spatial footprint of the action
- Use intervention-aware field to score candidate actions by expected future reduction per unit cost

Key idea:

The planner should choose actions based on which actions are expected to be most effective in reducing value-weighted risk

Experimental design and evaluation

Baselines:

- Reactive
- Prediction-only
- Proposed

Metrics:

- Value-weighted exposure
- Mean response time
- Energy or travel cost
- Task efficiency
- Communication overhead

Current Implementation Status

- A simulation framework for multi-robot bird deterrence in vineyard-like environments.
- The current codebase includes:
 - An online intensity model
 - Decentralized zone coordination
 - Candidate generation from hotspots
 - Benefit-vs-cost action scoring
 - Closed-loop intervention feedback
 - Experiment runners for thesis evaluation
- The staged thesis workflow already defines calibration, field-divergence checks, assignment tuning, method comparison, and robot scaling experiments

Closed-Loop Architecture of the Intervention Aware Multi-Robot Bird Deterrence System

